

Lokiva Group Hub

Collaborative Group Travel Synthesis, Real-Time Conflict Mediation, and Autonomous Itinerary Solver

Status: Shipped & Synced
Repository: Team-Vexoria/LOKIVA
Branch: main (Up to date)
Build: Passing (0 errors)
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1 The Problem: The WhatsApp Group Trip Planning Paradox

Group travel planning across India is notorious for high excitement followed by complete paralysis. More than 70% of planned group trips fail to materialize due to unstructured WhatsApp chatter, unexpressed financial anxiety, and incompatible vacation styles.

🔒 The Asymmetrical Budget Taboo

Friends rarely share their private spending ceilings openly. Low-budget travelers fear embarrassment and quietly drop out, while higher-budget travelers unwittingly propose stays or activities that stretch friends beyond their financial comfort.

🌍 Incompatible Cultural & Pace Tastes

One friend seeks high-energy Himalayan ridge trails, another wants slow riverside cafes and artisan handlooms, and a third expects royal fort photography. Standard booking apps only cater to single users, leaving groups in perpetual debate.

📱 The 500-Message WhatsApp Deadlock

Ideas get buried in WhatsApp threads, Instagram reels, and fragmented polls. Without a structured synthesis mechanism, decision fatigue sets in and groups default to inaction ("Let's plan next month").

⚡ Fragmented Execution & Itinerary Drift

Even when a city is chosen, planning daily logistics causes friction. Allocating stops, managing transit buffer times, and calculating fair per-person expense splits becomes a manual headache for the trip organizer.

Lokiva Core Mission

Replace chaotic WhatsApp travel planning with an intelligent, collective 60/40 workspace. One person creates a Group Room, invites friends with a link, each traveler completes a 60-second private preference quiz, and Lokiva autonomously synthesizes everyone's constraints into a harmonious consensus plan.

2 What We Have Built: 3-Phase Engineering Blueprint

We engineered **Lokiva Group Hub** directly into the LOKIVA core platform across three tightly integrated phases:

Phase	Delivered Modules	Core Capabilities
Phase 1: Entry, Auth & Quiz	- Hero CTA & Section Banner - RequireAuth Route Guard - Host Room Creator (/group/new) 5-Step Member Preference Quiz	Protected access ensuring only authenticated users enter room hubs. Members complete private budget, trip duration, pace, and cultural passion profiles in 60 seconds.
Phase 2: 60% Board & Sweet-Spot	- groupRecommendationEngine.ts - CrewSynergyRadar.tsx - GroupRecommendationBoard.tsx - Sweet-Spot Budget Equalizer	Deterministic multi-factor synthesis across 20+ Indian hubs. Calculates consensus score without hardcoding winners. Displays exact member attribution pills and financial fit tags.
Phase 3: 40% Chat & Solver Handoff	- GroupRealTimeChat.tsx - Card-to-Chat Drops & Live Reactions @Lokiva AI Conflict Mediator - Host Lock-In to /itinerary Solver	Full-height sticky real-time chat with cross-tab broadcast synchronization. Dropped cards with live Must Go/Maybe/Pass voting. Intelligent mediator suggesting split morning tracks and collective evenings. 1-click lock-in handoff into solver.

3 Core Engineering Innovations & Algorithms

The Budget Sweet-Spot Equalizer

To prevent low-budget exclusion, the engine computes a weighted group comfort ceiling that protects lower budgets while honoring collective possibilities:

$$\text{SweetSpot} = \text{round}(0.65 \times B_{\text{min}} + 0.35 \times B_{\text{median}})$$

Every destination card is labeled with financial transparency without exposing raw numbers:

✓ Fits All 4/4 Budgets Comfortably

⚡ Fits 3/4 Budgets · Stretches 1 by ~₹2,500

@Lokiva AI Conflict Mediator

Triggered when members type @lokiva or click the header button. The mediator compares divergent interests and generates contextual compromise blueprints:

"Harshit prioritizes mountain treks while Sneha favors artisan crafts, and the group sweet-spot is ₹21,000/person. **Dharamshala & Bir** is your 94% compromise: parallel morning trails (Triund hike vs. Norbulingka weaving studios) reuniting at sunset for Himalayan hearth dinners within budget."

4 The 60/40 Collaborative Workspace Architecture

Left 60%: Collaborative Intelligence Board

- ✦ **Crew Readiness Radar:** Displays all joined members, readiness pills (✓ Ready vs ⏳ Answering), and empty dashed waiting slots.
- ✦ **Simulate Friend Injector:** Allows one-click injection of realistic demo friends (Sneha, Aarav, Rohan, Ananya) to test multi-user synthesis on localhost.
- ✦ **Filter Tabs:** Top Group Consensus, Recommended For Me, and individual member filter chips (e.g. Harshit's Picks).
- ✦ **Exact Member Attribution:** Every card highlights who this place is recommended for (e.g. Recommended for 3 of 4: [✦ Harshit (You)] [✦ Sneha]).
- ✦ **Interactive Card Actions:** 🔥 Upvote (N), ❌ Pass/Veto, 💬 Drop to Chat, and 🔒 Lock Destination.

Right 40%: Real-Time Decision & Chat Dock

- ✦ **Multi-Tab Broadcast Sync:** Instant communication using BroadcastChannel('lokiva_group_sync') with zero page reloads.
- ✦ **Sender Visual Triad:** Right-aligned terracotta bubbles for the current user, left-aligned ivory bubbles for peers, and warm amber cards for AI Mediator.
- ✦ **One-Tap Card Drops:** Dropping a place renders an embedded mini-card in the chat stream with live reaction pills (🔥 Must Go, 🤔 Maybe, ❌ Pass).
- ✦ **Quick Prompt Chips:** One-tap shortcuts for @lokiva resolve our tie and Check our budget fit.
- ✦ **WhatsApp Crew Sharing:** One-tap invite link generator with pre-formatted WhatsApp invite message.

5 1-Click Host Lock-In to /itinerary Solver

When the group reaches consensus, the Host clicks 🔒 **Lock Winner & Build Group Itinerary** → (regular members have a ⚡ **Nudge Host** button). This triggers an automated handoff into the Lokiva itinerary engine:

Passed Parameters to /itinerary

/itinerary?city=Jaipur&groupId=grp-8492&travelers=4&budget=96000&perPersonBudget=24000&days=4&interests=crafts,food,forts&groupMode=true

Group Trip Mode Banner on /itinerary

Upon arrival at /itinerary, the solver displays a prominent "Group Trip Mode · Cost Split Per Person" banner showing:

- **Total Group Spend:** ₹96,000 for 4 travelers | **Exact Per-Person Split:** ₹24,000 / traveler
- **Fair Sweet-Spot:** Calibrated to protect lowest member ceiling without financial stretch
- **Navigation:** Direct link back to the Group Hub room.

6 Design Compliance & Repository Status

Sandstone Palette

Adheres strictly to zero dark fills on cards. Built with Sandstone Ivory (#FAF7F2), crisp ink typography (#12213B), terracotta (#C85A32), and warm borders (#E5DFD5).

Figma Typography Triad

Josefin Sans for display titles, Raleway for section titles & action buttons, Nunito for chat & body copy, JetBrains Mono for prices & telemetry. Zero serif fonts.

Git & Build Status

Fully pushed to Team-Vexoria/LOKIVA on branch main. Production build npm run build (tsc && vite build) passes in 8.07s with 0 errors.